```
student-22@VJTI-COMPs-iMac ~ % /usr/bin/python3 /Users/student-22/Desktop/blood_bank.py
```

VJTI BLOOD BANK MANAGEMENT SYSTEM
USE CASE: Process Blood Request
Automated Test Suite — White Box + Black Box
Student: Jiya Parekh | ID: 241071052

WHITE BOX TESTS — Statement + Branch + Path Coverage (WB_01 – WB_15)

ID	TEST NAME	INPUT SUMMARY	EXPECTED	ACTUAL	RESULT
WB_01	Path: Full Success	hosp=H101, bg=O+, u=2, stock=10	SUCCESS + analytics updated	SUCCESS + analytics updated	PASS
WB_02	Branch: Shortage Check	hosp=H101, bg=O+, u=10	DENIED (BloodShortageExcept…	DENIED (BloodShortageExceptio…	PASS
WB_03	Branch: Expiry Check	hosp=H101, bg=A+, u=1	DENIED (BloodExpiredExcepti…	DENIED (BloodExpiredException)	PASS
WB_04	Branch: Testing Status	hosp=H101, bg=B+, u=1	DENIED (BloodNotTestedError)	DENIED (BloodNotTestedError)	PASS
WB_05	Branch: Database Error	hosp=H101, bg=O+, u=2	DENIED (AnalyticsUpdateErro…	DENIED (AnalyticsUpdateError)	PASS
WB_06	Decision: Hospital Auth	hosp=H999, bg=O+, u=1	DENIED (UnauthorizedHospita…	DENIED (UnauthorizedHospitalE…	PASS
WB_07	Path: Multi-Batch Loop	hosp=H101, bg=O+, u=5, batchA=3+bat…	SUCCESS (multi-batch loop)	SUCCESS → SUCCESS	PASS
WB_08	Branch: Max Stock Limit	hosp=H101, bg=O+, u=1000	DENIED (ExceedsSystemLimitE…	DENIED (ExceedsSystemLimitErr…	PASS
WB_09	Statement: Signal Sender	hosp=H999 (invalid), bg=O+, u=1	denied_sent = True	denied_sent = True	PASS
WB_10	Statement: Analytics Update	hosp=H101, bg=O+, u=1, initial=10	analytics = 9 after dispatch	analytics = 9	PASS
WB_11	Branch: Null Request	NULL request	DENIED (NullRequestExceptio…	DENIED (NullRequestException)	PASS
WB_12	Decision: Rh Factor Match	hosp=H101, bg=A+, u=1	DENIED (BloodShortageExcept…	DENIED (BloodShortageExceptio…	PASS
WB_13	Path: System Maintenance	hosp=H101, bg=O+, u=2	DENIED (SystemLockedExcepti…	DENIED (SystemLockedException)	PASS
WB_14	Statement: Message Log	hosp=H101, bg=O+, u=1 (valid)	logMessageSent() called	log_contains_release = True	PASS
WB_15	Branch: Concurrency	2 threads, stock = 1 unit of O+	1 SUCCESS + 1 DENIED	1 success, 1 denied	PASS

White Box Summary → Total: 15 | Passed: 15 | Failed: 0 | Score: 100.0% [▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰]

BLACK BOX TESTS — ECP + BVA + Pre/Post Conditions (BB_01 – BB_15)

ID	TEST NAME	INPUT SUMMARY	EXPECTED	ACTUAL	RESULT
BB_01	Success Path (Main Flow)	bg=O+, u=2, stock=10, pretested	SUCCESS + analytics = 8	SUCCESS + analytics = 8	PASS
BB_02	Alt Flow 1 – Shortage	hosp=H101, bg=A+, u=15	DENIED (BloodShortageExcept…	DENIED (BloodShortageExceptio…	PASS
BB_03	Alt Flow 1 – Expired	hosp=H101, bg=B+, u=1	DENIED (BloodExpiredExcepti…	DENIED (BloodExpiredException)	PASS
BB_04	Alt Flow 2 – Not Tested	hosp=H101, bg=AB-, u=1	DENIED (BloodNotTestedError)	DENIED (BloodNotTestedError)	PASS
BB_05	BVA – Exact Stock	bg=O-, u=5, stock=5 (exact match)	SUCCESS + analytics = 0	SUCCESS + analytics = 0	PASS
BB_06	Invalid Hospital ID	hosp=H999, bg=O+, u=1	DENIED (UnauthorizedHospita…	DENIED (UnauthorizedHospitalE…	PASS
BB_07	Pre-cond: Empty Storage	hosp=H101, bg=A-, u=1	DENIED (BloodShortageExcept…	DENIED (BloodShortageExceptio…	PASS
BB_08	Post-cond: Analytics	bg=O+, u=2, initial stock=10	inventory = 8 post-dispatch	inventory = 8	PASS
BB_09	ECP – Negative Units	hosp=H101, bg=O+, u=-5	DENIED (InvalidUnitsExcepti…	DENIED (InvalidUnitsException)	PASS
BB_10	BVA – Zero Units	hosp=H101, bg=O+, u=0	DENIED (InvalidUnitsExcepti…	DENIED (InvalidUnitsException)	PASS
BB_11	Multiple Failures	bg=O+, u=20, stock=2 untested	DENIED (Shortage or NotTest…	DENIED (BloodNotTestedError)	PASS
BB_12	ECP – Special Chars in BG	hosp=H101, bg=O+!!, u=1	DENIED (InvalidBloodGroupEx…	DENIED (InvalidBloodGroupExce…	PASS
BB_13	Case Sensitivity Normalise	bg='ab+' (lowercase), u=2, stock=5	Normalised to AB+; SUCCESS	SUCCESS → SUCCESS	PASS
BB_14	Rapid Sequential Requests	Req1:u=5, Req2:u=6, stock=10	Req1 SUCCESS + Req2 DENIED	r1=OK, r2=DENIED	PASS
BB_15	Mandatory Field Check	hosp=H101, bg=, u=1	DENIED (InvalidBloodGroupEx…	DENIED (InvalidBloodGroupExce…	PASS

Black Box Summary → Total: 15 | Passed: 15 | Failed: 0 | Score: 100.0% [▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰]

COVERAGE REPORT

COVERAGE TYPE	SCOPE	TEST IDs	STATUS
Node Coverage	N1 — N13 (13 nodes)	WB_01 — WB_15	100%
Branch Coverage	All 8 decision outcomes	WB_02–WB_06, WB_08, WB_11–WB_13	100%
Path Coverage	8 distinct paths	WB_01, WB_02, WB_03, WB_04, WB_07	100%
Loop Coverage	0-iter, 1-iter, multi-batch	WB_07, BB_14	100%
Condition Coverage	12 condition outcomes	WB_01 — WB_15	100%
ECP — Valid Classes	bg, units, hosp (valid)	BB_01, BB_05, BB_13	Covered
ECP — Invalid Classes	bg, units, hosp (invalid)	BB_06, BB_09–BB_12, BB_15, WB_11	Covered
BVA — Boundaries	units = 0, -1, exact, max+1	BB_09, BB_10, BB_05, WB_08	Covered

OVERALL SUMMARY

Total Tests : 30
Passed : 30
Failed : 0
Score : 100.0% [▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰]